

Agentic AI for genetic risk prediction and reasoning in precision medicine

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Abstract

Polygenic risk scores (PRS) have become essential tools for quantifying genetic risk of complex diseases and are increasingly applied in clinical research for many chronic conditions such as cardiovascular disease, cancers, and brain disorders. However, the widespread adoption of PRS in precision medicine remains limited by two key challenges: (i) the absence of scalable automated systems for implementing PRS models across diverse populations and traits, and (ii) a lack of interpretability tools that can explain the biological and clinical relevance of risk predictions. We propose to develop a large language model-powered multi-agent AI system for interpretable PRS prediction and reasoning in precision medicine, integrated within our recently developed PennPRS platform¹ (<https://pennprs.org/>), the first cloud computing platform for scalable privacy-preserving PRS model training and application. Our agentic system will use model context protocol and retrieval-augmented generation to coordinate agents responsible for genetic prediction, scientific reasoning, and report generation. Outputs will include not only risk estimates but also transparent, literature-grounded explanations linking predictions to causal genes, biological pathways, and modifiable risk factors. We will validate our system across diverse populations and phenotypes using data from the UK Biobank, FinnGen, Million Veteran Program, and All of Us Research Program. This project will advance the state of agentic AI in genetic risk prediction, producing a public-facing platform that empowers researchers and clinicians with interpretable and actionable genetic risk insights.

Keywords – Agentic AI, Genetic risk prediction.

Introduction

Polygenic risk score (PRS) is a transformative tool in precision medicine, offering comprehensive predictions of an individual's genetic predisposition to complex diseases²⁻⁶. Advances in PRS methodology, the growing availability of rich genome-wide association studies (GWAS) data sources, and decreasing genetic testing costs are expected to continually enhance the utility and accessibility of PRS-based genetic risk prediction. For example, the GWAS Catalog⁷ has been curating and harmonizing GWAS summary statistics for over 90,000 phenotypes. These summary datasets, derived from numerous large-scale studies, offer valuable opportunities for us to conduct large-scale PRS model training and efficient PRS applications, which are increasingly crucial for personalized preventive healthcare and targeted early interventions^{6,8,9}.

Despite these advancements, two fundamental computational challenges prevent PRS applications from achieving widespread clinical integration. First, there remains a lack of end-to-end automated scalable infrastructure for deploying and managing pre-trained PRS models in real-world settings. Second, most existing methods lack explainable reasoning capabilities to interpret the biological relevance of PRS predictions. Clinicians and researchers need systems that not only efficiently deliver risk scores for diseases of interest but also provide contextual understanding, such as identifying causal genes, underlying biological mechanisms, and

modifiable risk factors, in a transparent and evidence-based manner. These challenges underscore the limitations of the traditional PRS framework, which is becoming increasingly inadequate to fully address the diverse and growing demands of widespread PRS applications.

Agentic AI is an emerging and rapidly expanding field, developing systems capable of automated learning and scientific reasoning¹⁰, empowering biomedical research through collaborative agents that integrate computational models and tools within a domain-specific scientific context¹¹. While several agentic platforms have been successfully developed in genomic areas (e.g., SpatialAgent¹² and CellAgent¹³), **no agentic AI framework currently exists for PRS-based genetic risk prediction in precision medicine**. To address these gaps, this project aims to develop a large language model (LLM)-powered, multi-agent AI system that not only empowers PRS predictions but also interprets and reasons over the results, effectively tackling critical computational and interpretability limitations.

This system will be **fully integrated within PennPRS¹** (<https://pennprs.org/>), our centralized cloud computing platform **built on AWS** for privacy-preserving, coding-free PRS model training and application. Notably, PennPRS is the first cloud computing platform designed specifically for PRS, offering scalable tools for researchers to build, apply, and share models across institutions and cohorts. As the development team behind PennPRS, we have already generated and publicly released PRS models for over 8,000 phenotypes, supporting a growing user base. Our team also has extensive experience developing both single- and multi-ancestry PRS methods and applying them across large-scale biobank datasets including the UK Biobank, FinnGen, the Million Veteran Program (MVP), and All of Us¹⁴⁻¹⁹. This proposal represents the **next phase of PennPRS**: advancing from predictive modeling to end-to-end automated application and intelligent interpretation. We aim to build an agentic AI system that performs genetic risk prediction and stratification for complex diseases, interprets the scientific meaning of results, and produces clear, user-friendly reports for downstream use in research and clinical care.

Methods

Our system will follow a modular, multi-agent design, orchestrated using the Model Context Protocol (MCP) to support transparent agent communication and reasoning traceability. The architecture will include three core components: (i) a Prediction Agent, (ii) a Reasoning Agent, and (iii) a Report Generator, all integrated into the PennPRS cloud computing platform.

The Prediction Agent will automate the application of our pre-trained PRS models to user-provided individual-level genotype data for a specified disease of interest (e.g., cancers). It will generate predicted disease risk using survival models (e.g., risk estimates at different ages) and report results by referencing a representative cohort, for example, projecting that an individual's placement would be in the top 10% risk group at age 60, with a genetic risk 5 (95% CI: 4.3 – 6.0) times higher than the average risk of the age group. This agent will support thousands of phenotypes for which we already have pre-trained PRS models¹, operating across multiple ancestry groups and disease domains, leveraging data from the UK Biobank, FinnGen, and MVP. To ensure robustness and generalizability, we will rigorously test and refine the framework using hold-out data from the All of Us project. We will also improve our algorithm by meta-prediction with additional important sociodemographic factors and clinical biomarkers²⁰. This agent will be fully integrated within our existing PennPRS web infrastructure.

The Reasoning Agent will leverage LLMs fine-tuned on biomedical literature and knowledge bases^{21,22} to interpret the biological and clinical implications of PRS results. This agent will use retrieval-augmented generation (RAG) to connect model outputs with curated resources such as KEGG, DisGeNET, Reactome, Open Target, ClinVar, Human Cell Atlas, and the Human Protein Atlas. In the last two decades, advances in statistical genomics and computational biology have yielded rich insights into the genetic architecture of complex traits and diseases, ranging from

associated loci and biological pathways to gene and protein expression at tissue and cell levels. Using these data resources, this agent will identify likely causal genes, responsible tissues and cell types, implicated pathways, shared genetic architecture with other phenotypes, and modifiable environmental factors, grounding its reasoning in the scientific literature.

The Report Generator will synthesize the outputs of the prediction and reasoning agents into structured, human-readable reports. These reports will be tailored to researchers and clinicians, including risk estimates, biological narratives, references, and interpretability annotations. By using MCP to coordinate agent communication, the system will retain transparent logs of decision-making steps, increasing both trust and auditability. We will also conduct large-scale applications of our framework across thousands of phenotypes in the All of Us database and perform human evaluations of the generated reports in collaboration with clinical researchers at Penn and experts in the PRS and precision medicine research communities. The agentic system will be implemented using open-source LLMs such as LLaMA. Knowledge indexing and retrieval will use tools such as Haystack, LangChain, and FAISS, and the system will be deployed as Docker-based microservices for reproducibility and scalability.

Expected results

This project will deliver an integrated multi-agent AI system for genetic risk prediction and interpretation, deployed within the PennPRS cloud platform¹. Over the first three months, we will finalize the system architecture, build the communication layer using MCP, and prepare a harmonized set of evaluation datasets across UK Biobank, FinnGen, MVP, and All of Us. From months four to six, we will integrate the Prediction Agent and automate PRS prediction pipelines. Between months seven and nine, we will build and validate the Reasoning Agent, establishing a high-quality RAG pipeline for biomedical knowledge integration. By month twelve, we expect to complete the Report Generator and release the first open-source version of the full multi-agent system, along with tutorials, example reports, and API documentation.

We will submit our first technical report and results by the end of year one and present at relevant conferences such as NeurIPS (AI4Science) or ASHG. From months twelve to eighteen, we will apply the agents to diverse phenotypes using All of Us data and prepare manuscripts for submission to journals such as *Nature Medicine* or *Nature Genetics*. The PI and co-PI have strong record of publications in top scientific journals²³⁻²⁷. The final six months will focus on generalization across more traits and ancestries and preparation for broader public deployment.

This project will demonstrate how AI agents can advance the clinical utility and interpretability of PRS-based genetic risk prediction models more broadly. The final deliverable will be integrated into our PennPRS platform, an open cloud-based platform that empowers the research community with interpretable, intelligent, and scientifically grounded prediction, paving the way for AI-assisted decision-making in genetic risk prediction and precision medicine.

Funds needed

We request \$69,000 in cash funding to support two student research assistants working on the development and evaluation of the proposed agentic AI framework within the PennPRS platform. This includes tasks such as model fine-tuning, infrastructure integration, and open-source tool release. We also request \$49,000 in AWS Promotional Credits to support key cloud services, including Amazon EC2, S3, SageMaker AI, and API Gateway, for training, deployment, and system scaling. The current PennPRS platform operates on AWS at an average cost of \$1,000-\$2,000 per month, and these additional resources are essential to support the expanded functionality of the integrated PennPRS ecosystem.

Additional information

1. Open-source tools we plan to contribute to: open-source AI agents, integrated within PennPRS.
2. AWS ML tools we will use: SageMaker AI, Amazon EC2, and Step Functions.

Appendix A

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